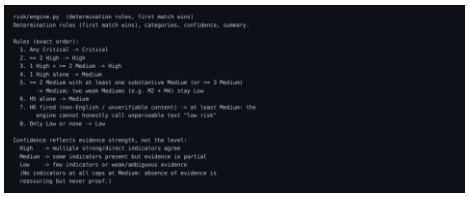
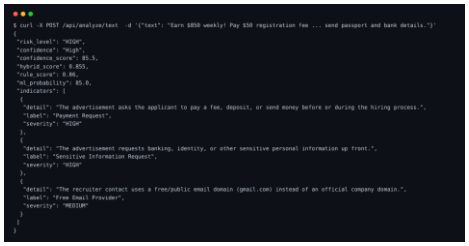
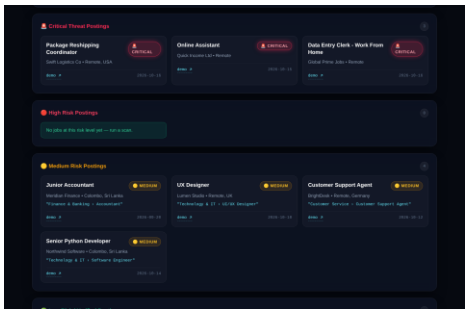
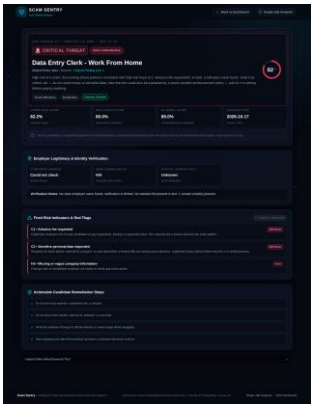
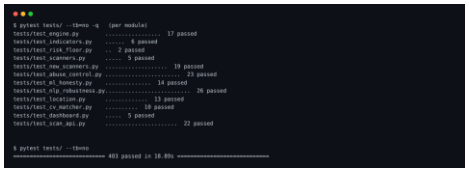
ESSENTIALS OF ARTIFICIAL INTELLIGENCE

INDIVIDUAL CONTRIBUTION SHEET

Student Name	Yasith Nirmal Peiris	**Registration No.**	**D/BSE/25/0018**
Degree / Intake	Information Technology / 2024	**Group No.**	44
Project Title	Scam Sentry: An Intelligent Fake Job Advertisement Detection System		

Technical Contributions:

Task / Component	My Contribution	Tools / AI Technique Used	Evidence / Output
1. Risk Engine & indicators	Designed the rule-based Risk Engine: graded indicators (C1-C4 critical, H1-H6 high, M1-M5 medium, L1-L3 low), first-match-wins determination rules, evidence-based confidence, language discipline (never claims 'fake'), employer verification, freshness / ghost-job signals and a deterministic safety net so a Critical indicator can never be averaged away by the ML score.	Rule-based reasoning Python pytest (test_engine, test_risk_floor)	 *Fig 1 - risk/engine.py determination rules*  *Fig 2 - Analysis API response*
2. Job board scanner & scheduler	Built the scanner framework with a plug-in registry and ten public-source scanners (RemoteOK, Remotive, Arbeitnow, Jobicy, Himalayas, Working Nomads, LaraJobs, Greenhouse, Lever, Ashby), URL de-duplication, per-source error isolation and the scheduled scan service that runs every listing through the Risk Engine.	Python, requests Python threading scheduler Public JSON / RSS APIs	 *Fig 3 - Scanned jobs grouped by risk level*

3. Storage, security & API	Implemented the storage layer (SQLite and PostgreSQL behind one API, location backfill) and the REST endpoints for scanning, jobs, history and sources. Added scan-token auth, per-IP rate limits, CSP headers and fail-closed LLM start-up; added the optional LLM explanation layer.	Flask, Flask-Limiter SQLite / PostgreSQL Render deployment	 *Fig 4 - Job dossier served from stored data*
4. ML calibration, taxonomy, location & CV matching	Calibrated the ML probability and labelled it honestly as a supporting signal trained on synthetic data. Built the industry / job-role taxonomy, country and region resolution (including bare city names), dashboard filters and the CV matcher that ranks jobs by relevance, never as a risk verdict.	scikit-learn calibration Python, JavaScript Keyword / taxonomy matching	 *Fig 5 - Commit history (git log)*
5. Testing & CI	Wrote the large automated test suite (indicators, risk floor, scanners, abuse control, NLP robustness, location, CV matcher, dashboard, scan API) and CI workflow, and recreated the project documentation. All 403 tests pass.	pytest GitHub Actions Python	 *Fig 6 - pytest results (403 passed)*

Declaration

I confirm that the contributions listed above accurately represent my individual work in this group project, and I can explain or demonstrate the stated work if required.

Student Signature: _____ **Date:** 17/10/2026

Evidence may include source code, model/notebook, diagram, dataset work, test results, screenshots, report section or presentation slides.